

Advanced Engineering Mathematics

Complex Numbers

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1 Complex Numbers

Key Concept

A **complex number** is written in rectangular form as

$$z = x + iy,$$

where $x, y \in \mathbb{R}$ and $i^2 = -1$.

The quantities

$$\operatorname{Re}(z) = x, \quad \operatorname{Im}(z) = y$$

are called the **real part** and **imaginary part** of z , respectively.

The **complex conjugate** of z is

$$z^* = x - iy,$$

and the **modulus** of z is

$$|z| = \sqrt{x^2 + y^2}.$$

Complex numbers extend the real number system in a natural way. In earlier engineering mathematics, variables were often real-valued. Now the independent or dependent variable may be complex. This is important because many engineering tools, especially the Laplace transform and the Z -transform, use complex variables.

A useful viewpoint is geometric: the number

$$z = x + iy$$

is represented by the point (x, y) in the complex plane. The horizontal axis is the real axis, and the vertical axis is the imaginary axis.

Key Concept

For

$$z_1 = x_1 + iy_1, \quad z_2 = x_2 + iy_2,$$

the basic operations are

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2),$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2),$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1),$$

and if $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2}.$$

Also,

$$zz^* = |z|^2 = x^2 + y^2, \quad \operatorname{Re}(z) = \frac{z + z^*}{2}, \quad \operatorname{Im}(z) = \frac{z - z^*}{2i}.$$

Motivation and Intuition

Why do we need complex numbers in engineering? A sinusoid such as

$$A \cos(\omega t + \phi)$$

can be handled much more efficiently through

$$Ae^{i(\omega t + \phi)}.$$

Instead of separating magnitude and phase repeatedly, complex numbers allow both to be carried at once in a single expression.

Geometric Interpretation

If $z = x + iy$, then:

- $|z|$ is the distance from the origin to the point z .
- z^* is the reflection of z across the real axis.
- Addition corresponds to vector addition in the plane.
- Multiplication has a deeper geometric meaning that becomes clearer in polar form.

Key Properties

Key Concept

Important modulus properties:

$$|z_1 z_2| = |z_1| |z_2|, \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (z_2 \neq 0),$$

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

(the triangle inequality).

These are frequently used in complex analysis, signal processing, and control.

Example

Example 1: Simplify a quotient

Simplify

$$\frac{3 - 2i}{-1 + i}.$$

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{3 - 2i}{-1 + i} \cdot \frac{-1 - i}{-1 - i} = \frac{(3 - 2i)(-1 - i)}{(-1 + i)(-1 - i)}.$$

Compute the numerator:

$$(3 - 2i)(-1 - i) = -3 - 3i + 2i + 2i^2.$$

Since $i^2 = -1$,

$$-3 - 3i + 2i + 2(-1) = -5 - i.$$

Compute the denominator:

$$(-1 + i)(-1 - i) = (-1)^2 - i^2 = 1 - (-1) = 2.$$

Hence,

$$\frac{3 - 2i}{-1 + i} = \frac{-5 - i}{2} = -\frac{5}{2} - \frac{1}{2}i.$$

Example

Example 2: Use the conjugate to find real and imaginary parts

Let

$$z = 4 - 3i.$$

Then

$$z^* = 4 + 3i.$$

Now,

$$\operatorname{Re}(z) = \frac{z + z^*}{2} = \frac{(4 - 3i) + (4 + 3i)}{2} = \frac{8}{2} = 4,$$

and

$$\operatorname{Im}(z) = \frac{z - z^*}{2i} = \frac{(4 - 3i) - (4 + 3i)}{2i} = \frac{-6i}{2i} = -3.$$

Engineering Note

In AC circuit analysis, impedance is written as a complex number:

$$Z = R + iX,$$

where R is resistance and X is reactance. This compact form allows algebraic handling of phase shifts and amplitudes in one framework.

Common Mistakes / Notes

- Do not confuse z^2 with $|z|^2$. In general,

$$z^2 = (x + iy)^2 = x^2 - y^2 + i2xy,$$

whereas

$$|z|^2 = x^2 + y^2.$$

- The imaginary part of $z = x + iy$ is y , not iy .
- When dividing complex numbers, always rationalize the denominator using the conjugate.